



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
In-Semester Examination I
Academic Year: 2025-26 (Semester I)

Class: TY	Subject: Manufacturing Processes - II	Subject Code: PCC-ME304
Date: 20- Aug-2025	Time: 15:00 to 15:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	Solve all the questions	Marks	CO	BT
Q.1	When is forced chip breaking necessary?	5	CO1	BT2
Q.2	What is rake angle? Explain their types with the help of a schematic diagram.	5	CO2	BT1
Q.3	Covert from ASA system to ORS by Graphical method.	5	CO2	BT1
Q.4	During turning a ductile alloy by a tool of $\gamma_o = 10^\circ$ it was found $P_z = 1000\text{N}$, $P_x = 400\text{ N}$, $P_y = 300\text{ N}$ and $\zeta = 2.5$. Evaluate, using MCD, the values of F , N and μ as well as P_s and P_n for the above machining.	5	CO1	BT3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Demonstrate the metal cutting process and select the appropriate cutting tools for it
CO 2	Exhibit the characteristic, terminologies related to cutting tools.
CO 3	Calculate the machining time, depth of cut, feed required.
CO 4	Demonstrate knowledge related to various machining processes.
CO 5	Exhibits the knowledge of various finishing processes.
CO 6	Select proper machining processes for the product to be manufacture



Class: TY B.Tech.

Course: Theory of Machines and Mechanisms

Course Code: PCC-ME303

Date: 20.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

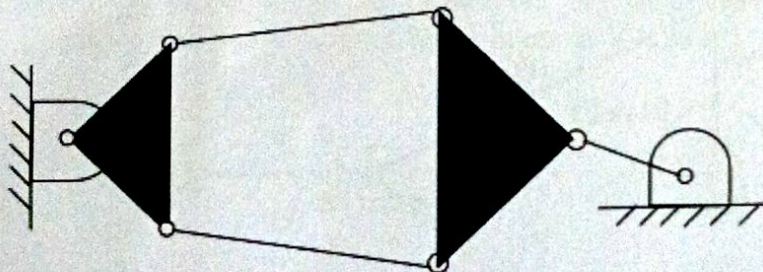
- i. *Solve Section A or Section B.*
- ii. *Solve Section C or Section D.*
- iii. *Section E is compulsory.*

Q.N.

Marks CO BT

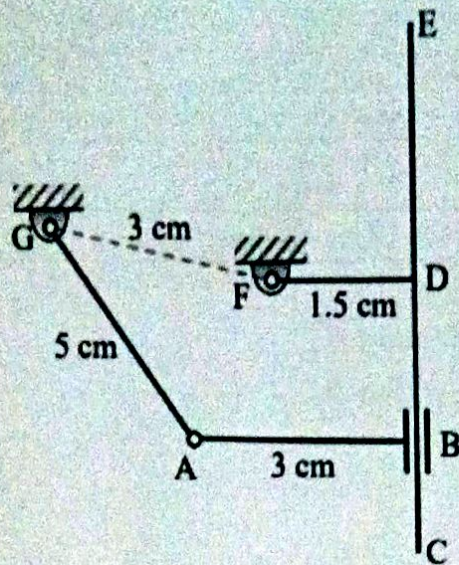
Section A

- | | | | | | |
|-----|---|---|----|-----|-------------|
| Q.1 | a | The number of degrees of freedom in a planar mechanism having n links and j simple hinge joints is... | 01 | Co1 | BT1 |
| | b | Distinguished kinematics pairs based on nature of contact. Also differentiate the Kinematic chain and kinematic mechanism | 03 | Co1 | BT1 |
| | c | The number of degrees of freedom of the linkage shown in the figure is | 02 | Co1 | BT2,
BT3 |



Section B

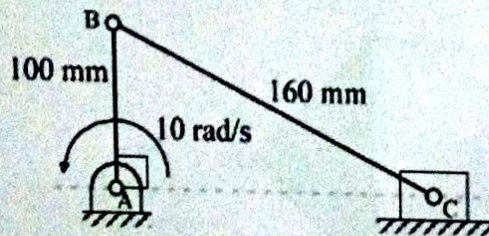
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|-----|---|---|----|-----|-------------|
| Q.1 | a | A planar mechanism has 8 links and 10 rotary joints. The number of degree of freedom of the mechanism, using Gruebler's criteria is... | 01 | Co1 | BT1 |
| | b | Explain inversion of single slider crank chain mechanism. | 03 | Co1 | BT1 |
| | c | The number of degree of freedom of a planar four-bar linkage mechanism with 3 revolute kinematic pairs and 1 prismatic kinematic pair is shown in the figure. | 02 | Co1 | BT2,
BT3 |



Section C

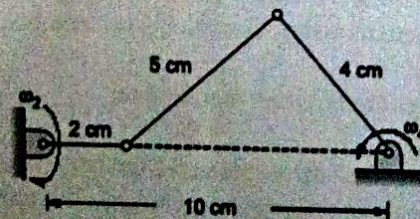
- 2 a In a slider-crank mechanism, the lengths of the crank and the connecting rod are 100mm and 160mm, respectively. The crank is rotating with an angular of 10 radian/s counter-clockwise. The magnitude of linear velocity (in m/s) of the piston at the instant corresponding to the configuration shown in the figure is ____.

04 Co2 BT3



- b In the configuration of the planar four-bar mechanism at a certain instant as shown in the figure, the angular velocity of the 2 cm long link is $\omega_2 = 5 \text{ rad/s}$. Given the dimensions as shown, the magnitude of the angular velocity ω_4 of the 4 cm long link is given by ____ rad/s (round off to 2 decimal places).

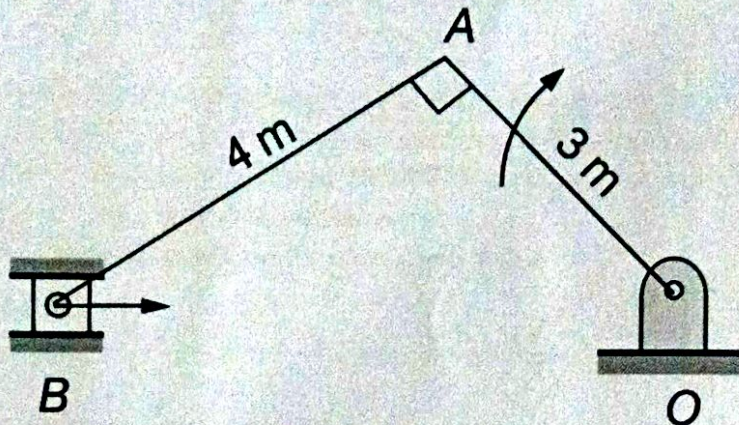
04 Co2 BT3



Section D

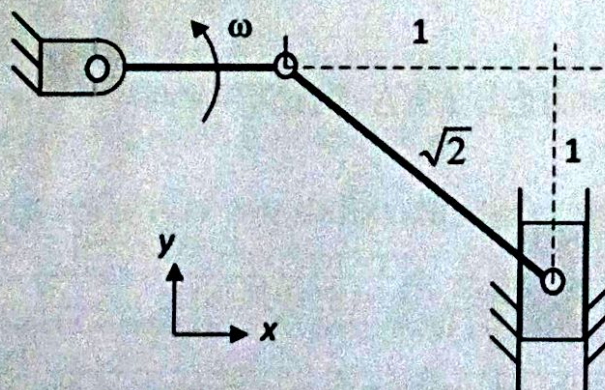
- 2 a For an inline slider-crank mechanism, the length of the crank and connecting rod are 3 m and 4 m, respectively. At the instant when the connecting rod is perpendicular to the crank, if the velocity of the slider is 1 m/s, the magnitude of angular velocity (upto 3 decimal points accuracy) of the crank is _____ radian/s.

04 Co2 BT3



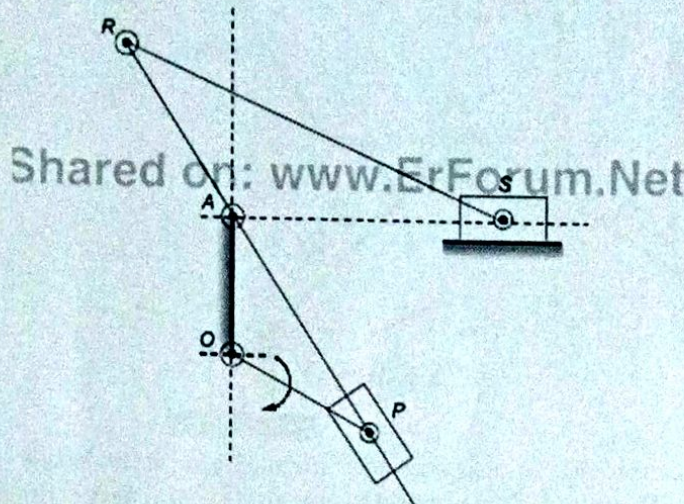
- b The crank of a slider-crank mechanism rotates counter-clockwise (CCW) with a constant angular velocity ω , as shown in fig. Assume the length of the crank to be r . find out the slider velocity inm/s. (find out velocity of Slider in terms of ω and r)

04 Co2 BT3



Section E

- 3 a The Whitworth quick return mechanism as shown in fig. with link length as follows. 03 Co1 BT2
 $OP = 300 \text{ mm}$, $OA = 150 \text{ mm}$, $AR = 160 \text{ mm}$, $RS = 450 \text{ mm}$.
 The Quick return ratio for the mechanism is.....



- b Find the minimum and maximum transmission angle for the mechanism. Consider following length of different link. 03 Co1 BT2, BT3
 Fixed link = 30 mm
 Coupler link = 40 mm
 Crank link = 10 mm
 Rocker link = 30 mm

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Demonstrate the working principles of various commonly used mechanisms in mechanical systems.
CO 2	Construct and evaluate velocity and acceleration diagrams of mechanisms using graphical methods.
CO 3	Understand and analyze the motion of cams and followers, and design cam profiles for specific follower motions.
CO 4	Explain the working and application of flywheels and governors, and evaluate their role in regulating speed fluctuations.
CO 5	Understand gear tooth geometry, select suitable types of gears based on design needs, and analyze their motion characteristics.
CO6	Analyze simple and compound gear trains used in machines for speed and torque transmission, considering their efficiency and application.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
In-Semester Examination I
Academic Year: 2025-26 (Semester – I)

Class: TY B.Tech.	Subject: Introduction to Industry 4.0	Subject Code: PEC-ME-303
Date: 18.08.2025	Time: 15:00 to 15:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/sharing of stationery, calculator, etc. is not allowed.
- iv. Figures to the right indicate full marks.
- v. Solve **any four questions** from questions 1 to 5.

Q.N.		Marks	CO	BT
Q.1	What is the internet of services (IoS) and how does it support the infrastructure of IoT? Provide Examples of SaaS, SaaS, Digital Twins.	5	CO3	BT1,2
Q.2	Describe the progress of Industry 4.0 in the USA, Europe and China. Why has India not yet achieved similar advancement?	5	CO6	BT3
Q.3	How do drivers and enablers help to implement Industry 4.0 in today's automated world?	5	CO1	BT2
Q.4	Discuss the key features and major inventions of each Industrial revolution in Industry 4.0.	5	CO1	BT1,2
Q.5	A furniture manufacturer uses wood; metal parts and upholstery materials sourced from different suppliers. Delays in material delivery are causing missed customer deadlines. Explain how predictive analytics could forecast material shortages and prevent production delays.	5	CO2	BT2,3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the drivers and enablers of Industry 4.0.
CO 2	Appreciate the smartness in Smart Factories, Smart Cities, Smart Products and Smart Services.
CO 3	Identify the various system used in a manufacturing plant and their role in an Industry 4.0 world.
CO 4	Appreciate the power of Cloud Computing in a networked economy.
CO 5	Understand the opportunities and challenges brought about by Industry 4.0.
CO6	Analyze how organizations and individuals should prepare to reap the benefits of Industry 4.0.

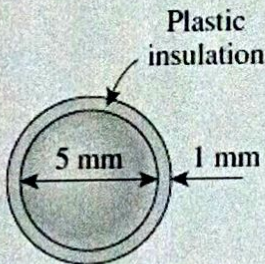


Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
In-Semester Examination I
Academic Year: 2025-26 (Semester -I)

Class: TY	Subject: Heat Transfer	Subject Code: PCC-ME-301
Date: 19.08.2025	Time: 15:00 to 15:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. This paper contains one page.

Q.N.		Marks	CO	BT
Q.1	Explain, using clear reasoning or an equation, how temperature affects the thermal conductivity of silver (a metal), asbestos (a non-metal), and hydrogen gas.	4	CO2	BT 1-2
Q.2	A 5-mm-diameter spherical ball at 50°C is covered by a 1-mm-thick plastic insulation ($k=0.13 \text{ W/m}\cdot\text{K}$). The ball is exposed to a medium at 15°C, with a convection heat transfer coefficient of $20 \text{ W/m}^2\cdot\text{K}$. Determine if the plastic insulation on the ball will increase or decrease heat transfer from the ball. 	4	CO1	BT 3-4
Q.3	A brick wall ($k=0.9 \text{ W/m}\cdot\text{K}$) of thickness 0.18 m separates the warm air in a room from the cold ambient air. On a particular winter day, the outside air temperature is -5°C and the room needs to be maintained at 27°C (inner wall temperature). The heat transfer coefficient associated with outside air is $20 \text{ W/m}^2\cdot\text{K}$. Neglecting the convective resistance of the air inside the room, find out the heat loss in (W/m^2).	6	CO 1-2	BT 3-4
Q.4	A coolant fluid at 30°C flows over a heated flat plate maintained at constant temperature of 230°C . The boundary layer temperature distribution at a given location on the plate may be approximated as $T=150+100\exp(-y)$ where y (in m) is the distance normal to the plate and T is in $^\circ\text{C}$. If thermal conductivity of the fluid is $1.0 \text{ W/m}\cdot\text{K}$, Determine the local convective heat transfer coefficient (in $\text{W/m}^2\text{K}$) at that location.	6	CO 1-2	BT 3-4

=== ALL THE BEST ===

Bloom's Taxonomy (BT) BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Student will be able to analyze heat conduction with and without internal heat generation, critical thickness of insulation and extended surfaces with the practical utilities.
CO 2	Student demonstrates the concept and mechanism of convection, conduction and radiation.
CO 3	Student exhibits application of mathematical approach to analyze and solve the numerical on complex heat transfer phenomena's.
CO 4	Student can explain the design, performance analysis and practical applications of heat exchangers.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
In Semester Examination -1
Academic Year: 2025-26 (Semester – I)

Class: TY	Subject: Modern Mobility/Internal Combustion Engines and E-Vehicles	Subject Code: PECME305/PEC-ME-311
Date: 21.08.2025	Time: 15:00 to 15:45	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.*
- The symbols carry the usual meaning.*
- This paper contains 1 page*

Q.N.		Marks	CO	BT
1.	Explain the working of two stroke engine with neat sketch	5	1	2
2.	Why is the stoichiometric mixture called chemically correct, and when does an engine need rich or lean mixtures? Explain with example	5	1	3
3.	With a neat sketch, explain the construction and working of a solid injection system in diesel engines.	5	2	2
4.	The following observations refer to a trial on a single cylinder diesel engine. Break Power = 75 kW, break thermal efficiency 35%, Mechanical efficiency 90%, Calorific value of diesel (CV) is 40 MJ/kg. Determine Indicated Power, Friction Power, Fuel consumption in kg/hr.	5	3	4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO1. Analyse IC engine components, working, and cycles CO2. Understand combustion processes and ignition systems in engines CO3. Evaluate engine performance and apply emission control strategies CO4. Assess electric vehicle technologies and global trends CO5. Optimize EV performance through motor and battery management CO6. Design hybrid powertrains and explore sustainable transportation



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Academic Year: 2025-26

Semester: I

Examination: ISE-I

Class: Third Year

Branch: Mechanical Engineering

Date: 22/08/2025

Time: 15:00 - 15:45

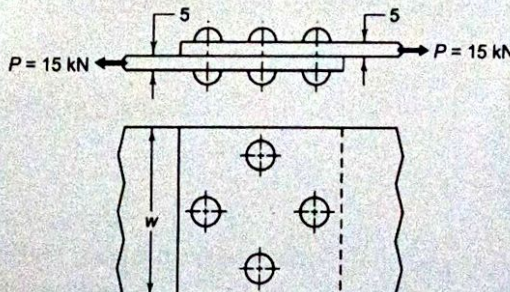
Subject: Machine Design - I

Code: PCC-ME305

Maximum Marks: 20

General Instructions:

- I. This paper contains 1 (One) page. Figures to right indicate full marks.
- II. Draw neat & compact sketches wherever necessary.
- III. Assume data wherever required and state them clearly.

Que. No.	Question	Marks	CO	BT
1.	<i>"strength is an inherent property of the material, whereas stress is not."</i> Elaborate the sentence as per your understanding.	5	CO1	BT2
2.	Consider you put heavy wet cloths to dry, which creates a tension of $0.1kN$ on the metal wire. After looking at you, mother shouted for not putting this much load on the wire to avoid its breaking. As an engineer immediately you measure the diameter of wire as 1 cm and identified the material having a ultimate strength of 500 kPa . Comment if your mother is correct or wrong, and why?	5	CO2	BT2
3.	Two plates, each 5 mm thick, are connected by means of four rivets as shown in figure. The permissible stresses for rivets and plates in tension, shear and compression are 80 , 60 and 120 N/mm^2 respectively. Calculate: i. Diameter of the rivets. ii. Width of the plate. 	5	CO2	BT5
4.	Identify the criterion of failure in the following aspects: i. A structure in the cross wind. ii. structure in tornado. iii. Diving board when swimmer is about to dive. iv. Screw during fitting. v. Wings of the airplane.	5	CO1	BT4

**** ALL THE BEST ****

- CO1: Analyze the fundamental principles of machine design to identify and evaluate the design requirements, material selection, and safety considerations for various mechanical components.
- CO2: Apply theories of static failure to design and analyze mechanical components subjected to static loading.
- CO3: Evaluate and design mechanical components to withstand fluctuating and fatigue loading conditions using endurance limit estimation.
- CO4: Design and analyze shafts, keys, and couplings as per industrial standards.
- CO5: Develop designs for threaded and welded joints subjected to static, eccentric, and fluctuating loads.
- CO6: Design helical springs to meet static and fatigue loading requirements.

BT1: Remember, BT2: Understand, BT3: Apply, BT4: Analyze, BT5: Evaluate, BT6: Create.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
In-Semester Examination I
Academic Year: 2025-26 (Semester – I)

Class: T.Y.	Subject: Design of Machine Elements	Subject Code: PCC-ME-303
Date: 22.08.2025	Time: 15:00 to 15:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/sharing of stationery, calculator, etc. is not allowed.
- iv. Figures to the right indicate full marks.
- v. The symbols carry the usual meaning.

Q.1	SOLVE.			
(a)	Draw S-N curve in terms of Low cycle and High cycle Fatigue.	Marks		
(b)	Explain the concept 'Design Consideration for Dynamic and Static Load.'	05	CO1	BT1
		05	CO1	BT2
Q.2	SOLVE.			
(a)	A cylindrical steel shaft is subjected to a fluctuating bending moment that varies from -100 Nm to +300 Nm. The shaft has a diameter of 20 mm. The material of the shaft is AISI 1045 steel, which has the following properties: ● Ultimate tensile strength (S_{ut}) = 580 MPa ● Yield strength (S_y) = 340 MPa Determine: 1. The factor of safety against fatigue failure for the shaft using the Soderberg criterion. 2. The modified endurance limit for the shaft, considering the following factors: a. Surface finish factor: 0.9 b. Size factor: 0.85 c. Reliability factor: 0.868 (for 99% reliability) d. Loading factor: 1.0 Assume: ● The shaft is subjected to bending stress only. ● The endurance limit for AISI 1045 steel (polished specimen) is $0.5 \times S_{ut}$. ● Use a factor of safety of 2 for design purposes.	10	CO2	BT5

Course Outcomes (CO)

Bloom's Taxonomy: BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

- CO1. Students understand the need, steps of machine design, and consideration for manufacturing
- CO2. Student will be able to evaluate life of machine components against static loads and fluctuating loads